CLOCK IN BY RADIANTS · SOLANA MOBILE HACKATHON · JUDGE ENCYCLOPEDIA

# Encyclopedia Elarion

## Defenders of the Realm: Echoes of Elarion

A depth brief for coaches: product, architecture, SKR, Orders, commerce, honesty — with sources.

Live on the Solana dApp Store · started May 2026 · **no revenue**

Companion to Technical Appendix (branch dev · c19c2a7c1 · read Oct 4). Pitch video: youtu.be/RUahALTRU7g

HOW TO READ THIS ENCYCLOPEDIA

# Depth for coaches — not a 60-second pitch

**01–04** Product · loops · town progression

**05–06** Architecture · CDN · Command Center

**07–10** MWA / SKR · Heartbound · Orders / Vigil · Commerce

**11–12** Heart arc (chat-join) · Arena honesty

**13–14** Feature status table · Quantified build facts

**15–17** Traction (safe) · Limitations · The ask

**Honesty locks for this deck:** no “12M SKR,” no “201 stakers,” no stake-threshold “realm living” tree wake. Heart wake = first chat join. Season prizes = Planned. Numbers cite Technical Appendix / METRICS Oct 4 or are omitted.

# A real strategy game for Solana Seeker

- **Problem:** few skill-based games on Solana mobile; token rewards often favor luck or capital
- **Product:** Echoes of Elarion — Home / Adventure / Arena loops on Seeker + web
- **SKR role:** identity, Heartbound tiers, Order gates, Vigil cosmetics, Jupiter checkout — never custody
- **Shipped:** Solana dApp Store (releases Sep 7 & Sep 24; live v2026.10.01)
- **Business:** pre-revenue; payment rail live and server-verified

Orders / Vigil / SKR surfaces on Seeker and web — not on the Google Play build.




**Signed-in title (address blurred)**

Source: live Seeker build screenshots · pitch assets Oct 4

2 · THREE LOOPS

# Home → Adventure → Arena → home again

### 1 · Home

Protect your home

1. Build & upgrade
2. Defend waves
3. Earn resources
4. Improve defenses

### 2 · Adventure

Discover the next challenge

1. Prepare gear
2. Explore / journey
3. Earn XP
4. Unlock content

### 3 · Arena

Lead the attack

1. Scout rival base
2. Deploy hero + troops
3. Fight the snapshot
4. Review & retaliate

↻ Arena sends you home to prepare again. Each loop feeds the next.

Product framing locked for Clock In pitch · Oct 4

# Four towns: early → mid → late → radiant Vigil




Early · wood / dying Heart




Mid · build & place




Late · fortified walls




Radiant · Vigil canopy

**Heart arc (Play lock):** dying → **alive on first chat join** (Life essence) → radiant after Vigil dressing. **Not** a stake-threshold “realm living” event.

Frames: Seeker device captures + live build plates (compressed). Labels describe visual stage, not analytics.

# Five tiers — client never trusted on stake/price/rewards

### Unity client
Unity 6000.4.8f1 · 49 asmdefs
Strict MVVM (e.g. StructureCardVM)
Android APK · WebGL · Windows

⇄

### Vercel API
api/ · 93 routes · 71 libs
Wallet-signed requests
8 scheduled crons

⇄

### Postgres (Neon)
api/schema.sql
46 migrations
stakes, Orders, saves, arena

### Cloudflare R2 CDN
Addressables bundles
Android 234 · WebGL 249 · Win 305

⇇

### Command Center
tools/command-centre.ps1
live tunables · kill switches
gate / preview / promote / prove

⇄

### Solana mainnet
MWA · native SKR stake
Genesis Token · Jupiter · Squads

Source: Technical Appendix p.03 · branch dev c19c2a7c1 · counted from repo Oct 4

3B · LIVE OPS WITHOUT SHIPPING A BUILD

# CDN SWAP · TUNABLES · SERVER IS TRUTH

### ① CDN CONTENT SWAP

Addressables → r2-ship.ps1 push → verify ALL targets. Android: **373** catalog generations on disk.

### ② LIVE TUNABLES

Command Center → client_tunables → clients poll ~30s. **109** allow-listed knobs · ~150s turnaround.

### ③ SERVER IS TRUTH

SKU catalog, stake tiers, Order banners computed server-side. Fail-soft tunables (build defaults on error).

| Target | Bundles | Catalogs | Newest catalog |
|---|---|---|---|
| Android | 234 | 373 | 2026.10.04.397868 |
| WebGL | 249 | 46 | 2026.10.04.397868 |
| Windows | 305 | 57 | 2026.10.02.395461 |

Technical Appendix · Content pipeline + zero-downtime pages · R2_PARITY_OK Oct 4 02:43 CT

# Connect → read-only stake → snapshot → unlock → pay

| Step | Mechanism |
|---|---|
| **1 Connect** | Mobile Wallet Adapter; ed25519 over 5-min wallet-bound nonce (burned) |
| **2 Read** | Server getAccountInfo on native stake — client never trusted to report stake (PRODUCT RULE 7) |
| **3 Snapshot** | skr_stake_snapshots · 7 statuses · verified_at grace |
| **4 Tier** | Resonance score → **11-tier** Heartbound ladder (server-told) |
| **5 Unlock** | Cosmetics + Order create/join/vote gated on verified stake |
| **6 Pay** | Jupiter ExactOut; verify at commitment finalized |

### Never custody

Player SKR stays staked in the player's wallet. Server reads chain; client renders.

### Cache / refresh

**300 s** stake cache · **60 s** manual cooldown. Cooldown from StakeConfig, not hardcoded.

### Anti-whale

Log curve: 500k SKR ≠ 500× of 1k. **21** curve tests + **28** telemetry tests.

Technical Appendix · SKR and wallet + anti-whale pages · api/_lib/skr-staking.js · heartbound-resonance.test.js

4B · HEARTBOUND RESONANCE · 11 TIERS

# StakePower + TenurePower → server tier

StakePower = 1000 × log10(1 + Stake/250)
TenurePower = MIN(700, 150 × ln(1 + Pulses/5))
minHeartboundStake = 100 · increases ramp over pulses; decreases immediate

Client is **told** its tier. Flash-staking does not stick.

heartbound-resonance-config.json · test/heartbound-resonance.test.js (21 tests)

| Tier | Name | Min score | Tier | Name | Min score |
|---|---|---|---|---|---|
| 0 | Silent | 0 | 6 | Hearttouched | 1800 |
| 1 | Emberbound | 300 | 7 | Deep Resonance | 2150 |
| 2 | Rootbound | 600 | 8 | Heartforged | 2500 |
| 3 | Stonebound | 900 | 9 | Eternal Echo | 2850 |
| 4 | Echoing | 1200 | 10 | Heartbound | 3200 |
| 5 | Awakened | 1500 | | | |

# Stake-gated clans; tenure-weighted Vigil

**Order** = clan. One per wallet. Create / join / vote require verified SKR stake (clan-stake-gate.js). Failed/degraded stake read → 403 STAKE_READ_FAILED first.

**Vigil** = Σ (percent_staked × tenure_seconds). Tenure from Postgres. 5 Vigil tiers: Ember → Dawn (0 / 1d / 7d / 30d / 90d).

**48h banners** (Lantern / Tide / Gale) — cosmetic only, Vigil-weighted plurality. Turnout bands scale with Order size.

**Genesis Token** = badge for Order join, **not** the stake gate. **Squads vault signers** must each hold a Seeker Genesis Token. Today: PARTIAL 0 vaults registered.

PARTIAL **Orders / Vigil usage** at owner-test scale — do not pitch as live clan meta.




Heart with Vigil dressing (radiant stage)

Technical Appendix · Orders and the Vigil · api/_lib/clan-stake-gate.js · clan-vigil.js · schema clan_*

# Jupiter ExactOut — priced and verified on server

| Stage | Where |
|---|---|
| Server prices the SKU | POST /api/purchases/quote — server decides the number |
| Short on SKR? | JupiterSwapService.cs ExactOut quote; wallet opens only if quote returns |
| Settlement | /api/purchases/verify reads transfer at commitment **finalized** |
| Health | Cron */15 synthetic ExactOut; HTTP 200 with error payload = DOWN |

**852/852**

swap-health checks passed, last 30 days

live DB read (Backend), Oct 4

### Honesty

Payment rail is live. Studio is **pre-revenue** for this pitch — do not lead with purchase counts or ad dollars. Failures alert via jupiter-fail-alert.js.

Technical Appendix · Pay in SKR page

# Dying → alive on first chat join → radiant after Vigil

| 1 | 2 | 3 |
|---|---|---|
| **Dying** | **Alive** | **Radiant** |
| Pre-chat Heart still dying | First chat join · Life essence line | Vigil tier + banner dressing |

**REMOVED claim**

Do **not** say canopy wakes because total verified stake crosses a threshold (“the realm is living”). That stake-threshold path is **not** the Play Heart arc for this submission.

UI: Cherry welcome or one-time modal (not a toast) paired with canopy wake. Persists across sessions after first join.




Dying / early Heart




Radiant after Vigil

Owner/Link lock Oct 5 · supersedes older pitch VO at ~3:18 and appendix realm-config stake-threshold copy for Play messaging

# Real base layout — matchmaking still Partial

- **ArenaTwoBase** hero-led assault into a defender **snapshot** (layout, troops, hero, gear)
- Defender hero fights as boss; WritesLiveSave = false — your town save is never touched
- Clash-of-Clans hook: hit a real base, rebuild, retaliate

PARTIAL **Matchmaking / real rival lookup** — today a **seeded demo rival**. Live player matchmaking is Planned rollout.

PLANNED Season prize pool from staked $10K SKR Integration prize **yield only** (40 / 25 / 15 / 20% shared 4–10). No payouts made.




Arena prep




Hero-led fight

Lead Dev lock Oct 4 · commit c19c2a7c1 · Technical Appendix Arena page · ArenaTwoBaseSession.cs

8 · FEATURE STATUS

# Implemented · Partial · Planned

| Feature | Status | One-line proof |
|---|---|---|
| MWA sign-in | IMPLEMENTED | Wallet-signed sessions; ed25519 nonce |
| On-chain SKR stake read | IMPLEMENTED | Server read-only; PRODUCT RULE 7 |
| Stake snapshot → tier | IMPLEMENTED | skr_stake_snapshots · 11 Heartbound tiers |
| Anti-whale scoring curve | IMPLEMENTED | 21 curve + 28 telemetry tests |
| Jupiter ExactOut checkout | IMPLEMENTED | Quote → swap → finalize verify; 852/852 health |
| Orders stake-gated | IMPLEMENTED | create/join/vote; STAKE_REQUIRED / STAKE_READ_FAILED |
| Vigil weight + 48h banners | IMPLEMENTED | stake%×tenure; Lantern/Tide/Gale cosmetic |
| Heart · chat-join wake | IMPLEMENTED | Dying → alive on first chat join (Life essence) |
| Heart Vigil dressing | IMPLEMENTED | Radiant tier + banner on Heart tree |
| ArenaTwoBase hero-led | IMPLEMENTED | vs snapshot; WritesLiveSave false |
| dApp Store live | IMPLEMENTED | Releases Sep 7 & Sep 24 |
| 10 languages | IMPLEMENTED | 58,120 keys; 15 locale regressions |
| Regression suite | IMPLEMENTED | 1,155 Unity decls · 2,014 API cases (counted) |
| Arena matchmaking | PARTIAL | Seeded demo rival today |
| Orders / Vigil usage | PARTIAL | Owner-test scale |
| Shared Order vaults | PARTIAL | Genesis Token gate on vault signers; 0 vaults |
| Season prize from yield | PLANNED | $10K SKR prize staked → yield 40/25/15/20 |
| Google Play / iOS / Pi | PLANNED | “Coming to” — not live |
| Matchmaking rollout | PLANNED | Real rival lookup next |

Lead Dev lock Oct 4 (c19c2a7c1) + Technical Appendix counts + Owner Heart arc lock Oct 5

8B · QUANTIFIED BUILD FACTS · VERIFIED ONLY

# Counted from repo / appendix / Backend logs

| | | |
|---|---|---|
| **5,967**<br>commits all branches · first May 18, 2026 | **1,155**<br>Unity [Test]/[UnityTest] declarations | **2,014**<br>API test cases · 120 suites |
| **93 / 71 / 46**<br>API routes · libs · migrations | **110**<br>canonical JSON catalogs | **10**<br>locales · 58,120 string entries |
| **11**<br>Heartbound tiers (0–10) | **300s / 60s**<br>stake cache / manual cooldown | **852/852**<br>Jupiter health · 30 days (Backend Oct 4) |

Technical Appendix regression + repo stats + SKR wallet pages. Omitted this pass if unverified: product SKU count “27,” 12M SKR, 201 stakers.

# Wallets, play, installs — no SKR totals

| | | |
|---|---|---|
| **538**<br>Solana wallets connected | **178**<br>peak daily players, dApp launch week | **2,292**<br>waves defended · high wave 49 |
| **1,700+**<br>installs in two weeks on dApp Store | **95**<br>dApp Store reviews · avg 3.9/5 | **1,674**<br>commits in hackathon window Sep 8–Oct 4 |

**Deliberately omitted:** 12M SKR held / 201 verified stakers (Link form lock — not cited on coach materials without fresh Backend verify). No “visitors.” No revenue.

METRICS.md / Backend Oct 4 · publisher portal for installs/reviews · commit window from METRICS hackathon facts

# What is shipped vs what is still soft

| Topic | Status | Coach-facing wording |
|---|---|---|
| Arena rivals | PARTIAL | Seeded demo rival today; matchmaking Planned |
| Orders / Vigil social | PARTIAL | Owner-test scale — not a live clan economy yet |
| Shared Order vaults | PARTIAL | Genesis Token required for vault signers; 0 vaults |
| Season prizes | PLANNED | Yield-only from Integration prize stake; 40/25/15/20% |
| Google Play / iOS / Pi | PLANNED | “Coming to” — Seeker dApp Store is live |
| Heart wake | LOCK | Chat-join Life essence — not stake-threshold realm living |
| SKR totals on forms | OMIT | Do not claim 12M / 201 without fresh Backend verify |
| Revenue | NONE | Pre-revenue; rail live |

Aligned with Owner/Link Oct 5 honesty locks + Lead Dev Partial notes

10 · THE ASK

# Play it. Stress it. Help it last.

- Play Echoes of Elarion on the Solana dApp Store
- $10K SKR Integration prize → staked principal; **yield only** funds seasons (Planned)
- **Four Seekers. Four dedicated testers. Zero excuses.**
- Encyclopedia + Technical Appendix are the depth pack for coaches

GitHub: [repo link to be added]

Pitch video (unlisted): youtu.be/RUahALTRU7g

Started May 2026 · no revenue · Clock In by Radiants